

First-Order ODEs (Zill 2.2–2.5)

$\frac{dy}{dx} = g(x)h(y)$, $\int \frac{dy}{h(y)} = \int g(x) dx + C$ – separable, integrate both sides.

$\frac{dy}{dx} + P(x)y = Q(x)$, $\mu(x) = e^{\int P(x) dx}$, $y = \frac{1}{\mu(x)} \left[\int \mu(x)Q(x) dx + C \right]$ – linear, integrating factor μ .

$M dx + N dy = 0$ exact iff $M_y = N_x$; $f(x, y) = \int M dx + g(y) = C$, $f_y = N$ – potential fn. f , $f_x = M$.

$\mu(x) = e^{\int (M_y - N_x)/N dx}$ (x -only), $\mu(y) = e^{\int (N_x - M_y)/M dy}$ (y -only) – restores exactness for non-exact eqns.

$\frac{dy}{dx} = f(y/x)$, sub $y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$ – homogeneous, reduces to separable in v, x .

$\frac{dy}{dx} + P(x)y = Q(x)y^n$, sub $w = y^{1-n} \Rightarrow \frac{dw}{dx} + (1-n)P(x)w = (1-n)Q(x)$ – Bernoulli, linear in w .

Linear Higher-Order ODEs (Zill 3.1–3.6)

$y_2 = y_1(x) \int \frac{e^{-\int P dx}}{[y_1(x)]^2} dx$ – reduction of order, 2nd solution from known y_1 of $y'' + Py' + Qy = 0$.

$ay'' + by' + cy = 0 \Rightarrow am^2 + bm + c = 0$ – homogeneous, const. coeff. (sub $y = e^{mx}$). Three root cases:

Distinct real m_1, m_2 : $y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$.

Repeated $m_1 = m_2$: $y = c_1 e^{m_1 x} + c_2 x e^{m_1 x}$.

Complex $m = \alpha \pm \beta i$: $y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$.

n th-order, repeated roots: real m (mult. k) $\rightarrow e^{mx}, x e^{mx}, \dots, x^{k-1} e^{mx}$; complex pair (mult. k) $\rightarrow e^{\alpha x} x^j \cos \beta x, e^{\alpha x} x^j \sin \beta x, j = 0, \dots, k-1$.

Undetermined coeff. – trial y_p : const. $\rightarrow A$; poly. deg. $n \rightarrow A_n x^n + \dots + A_0$; $e^{ax} \rightarrow A e^{ax}$; $\cos \beta x / \sin \beta x \rightarrow A \cos \beta x + B \sin \beta x$; product $\rightarrow$ product of forms. Duplicates y_c ? Multiply guess by x (or x^2).

$y_c = c_1 y_1 + c_2 y_2$, $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$, $u_1' = \frac{-y_2 f(x)}{W}$, $u_2' = \frac{y_1 f(x)}{W}$, $y_p = u_1 y_1 + u_2 y_2$ – variation of parameters.

$ax^2 y'' + bxy' + cy = 0$, sub $y = x^m \Rightarrow am(m-1) + bm + c = 0$ – Cauchy-Euler, $x > 0$. Distinct: $y = c_1 x^{m_1} + c_2 x^{m_2}$; repeated: $y = c_1 x^{m_1} + c_2 x^{m_1} \ln x$; complex: $y = x^\alpha [c_1 \cos(\beta \ln x) + c_2 \sin(\beta \ln x)]$.

$y = y_c + y_p$ – general solution; y_c solves homogeneous eqn, y_p is any one particular solution.

Laplace Transforms (Zill 4.1–4.5)

$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t) dt$, $t \geq 0$ – converges for s large, f piecewise cont., exp. order.

$\mathcal{L}\{af(t) + bg(t)\} = aF(s) + bG(s)$ – linearity.

Laplace transform pairs:

$\mathcal{L}\{1\} = 1/s$, $s > 0$, $\mathcal{L}\{t\} = 1/s^2$, $s > 0$

$\mathcal{L}\{t^n\} = n!/s^{n+1}$, $n = 1, 2, 3, \dots$, $s > 0$

$\mathcal{L}\{e^{at}\} = 1/(s-a)$, $s > a$

$\mathcal{L}\{\sin kt\} = k/(s^2 + k^2)$, $\mathcal{L}\{\cos kt\} = s/(s^2 + k^2)$, $s > 0$

$\mathcal{L}\{\sinh kt\} = k/(s^2 - k^2)$, $\mathcal{L}\{\cosh kt\} = s/(s^2 - k^2)$, $s > |k|$

$\mathcal{L}\{t \sin kt\} = 2ks/(s^2 + k^2)^2$, $s > 0$

$\mathcal{L}\{t \cos kt\} = (s^2 - k^2)/(s^2 + k^2)^2$, $s > 0$

$\mathcal{L}\{e^{at} \sin kt\} = k/[(s-a)^2 + k^2]$, $s > a$

$\mathcal{L}\{e^{at} \cos kt\} = (s-a)/[(s-a)^2 + k^2]$, $s > a$

$\mathcal{L}\{e^{at} t^n\} = n!/(s-a)^{n+1}$, $s > a$

$\mathcal{L}\{\mathcal{U}(t-a)\} = e^{-as}/s$, $s > 0$, $a \geq 0$ – unit step

$\mathcal{L}\{\delta(t-a)\} = e^{-as}$, $a \geq 0$ – Dirac delta

$\mathcal{L}^{-1}\{aF(s) + bG(s)\} = a\mathcal{L}^{-1}\{F(s)\} + b\mathcal{L}^{-1}\{G(s)\}$ – inverse; partial fractions, invert term by term.

$\mathcal{L}\{f'(t)\} = sF(s) - f(0)$; $\mathcal{L}\{f''(t)\} = s^2 F(s) - sf(0) - f'(0)$; $\mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - s^{n-1}f(0) - \dots - f^{(n-1)}(0)$ – transform of derivatives.

$\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{F(s)}{s}$ – transform of an integral.

$\mathcal{L}\{e^{at} f(t)\} = F(s-a)$ – 1st translation thm.; shift, $s \rightarrow s-a$.

$\mathcal{U}(t-a) = 0$ ($t < a$), 1 ($t \geq a$); $\mathcal{L}\{f(t-a)\mathcal{U}(t-a)\} = e^{-as}F(s)$ – 2nd translation thm.; else $\mathcal{L}\{g(t)\mathcal{U}(t-a)\} = e^{-as}\mathcal{L}\{g(t+a)\}$.

$\mathcal{L}\{f(t)\} = \frac{1}{1-e^{-sT}} \int_0^T e^{-st} f(t) dt$ – periodic fn., $f(t+T) = f(t)$.

$\int_0^\infty f(t)\delta(t-a) dt = f(a)$; $\mathcal{L}\{\delta(t-a)\} = e^{-as}$, $a \geq 0$; $\mathcal{L}\{\delta(t)\} = 1$ – Dirac delta (unit impulse).

$(f * g)(t) = \int_0^t f(\tau)g(t-\tau) d\tau$, $\mathcal{L}\{f * g\} = F(s)G(s)$ – convolution theorem.

Quick Reference – First-Order ODE Method by Type

Separable ($dy/dx = g(x)h(y)$) – separate, integrate.

Linear ($y' + P(x)y = Q(x)$) – $\mu = e^{\int P dx}$.

Exact ($M dx + N dy = 0$, $M_y = N_x$) – potential f , $f_x = M$, $f_y = N$, $f(x, y) = C$.

Non-exact – integrating factor $\mu(x)$ or $\mu(y)$ to restore $M_y = N_x$.

Homogeneous ($dy/dx = f(y/x)$) – sub $y = vx$.

Bernoulli ($y' + Py = Qy^n$) – sub $w = y^{1-n}$.

Quick Reference – Root-Case Forms Side by Side

Const. coeff. $ay'' + by' + cy = 0$ ($am^2 + bm + c = 0$) vs. Cauchy-Euler $ax^2 y'' + bxy' + cy = 0$ ($am(m-1) + bm + c = 0$, sub $y = x^m$):

Distinct m_1, m_2 – $y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$ vs. $y = c_1 x^{m_1} + c_2 x^{m_2}$.

Repeated m_1 – $y = c_1 e^{m_1 x} + c_2 x e^{m_1 x}$ vs. $y = c_1 x^{m_1} + c_2 x^{m_1} \ln x$.

Complex $\alpha \pm \beta i$ – $y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$ vs. $y = x^\alpha [c_1 \cos(\beta \ln x) + c_2 \sin(\beta \ln x)]$.

Quick Reference – Solving an IVP by Laplace Transform

1) Apply $\mathcal{L}\{\cdot\}$: use $\mathcal{L}\{y^{(n)}\} = s^n Y(s) - s^{n-1}y(0) - \dots - y^{(n-1)}(0)$ to get an algebraic eqn. in $Y(s)$.

2) Solve algebraically for $Y(s)$.

3) Partial fractions to match table entries; shifting thms. for $e^{\pm as}$ factors, convolution thm. for a product $F(s)G(s)$.

4) Invert term by term: $y(t) = \mathcal{L}^{-1}\{Y(s)\}$.